

C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
<http://c-immsim.iac.rm.cnr.it> (alias to <http://kraken.iac.rm.cnr.it/C-IMMSIM>)

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
<https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262>

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
<https://jitc.bmj.com/content/9/5/e002694>

Original C-IMMSIM model: www.iac.cnr.it/~filippo/c-immsim

GETTING HELP:

Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)

Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

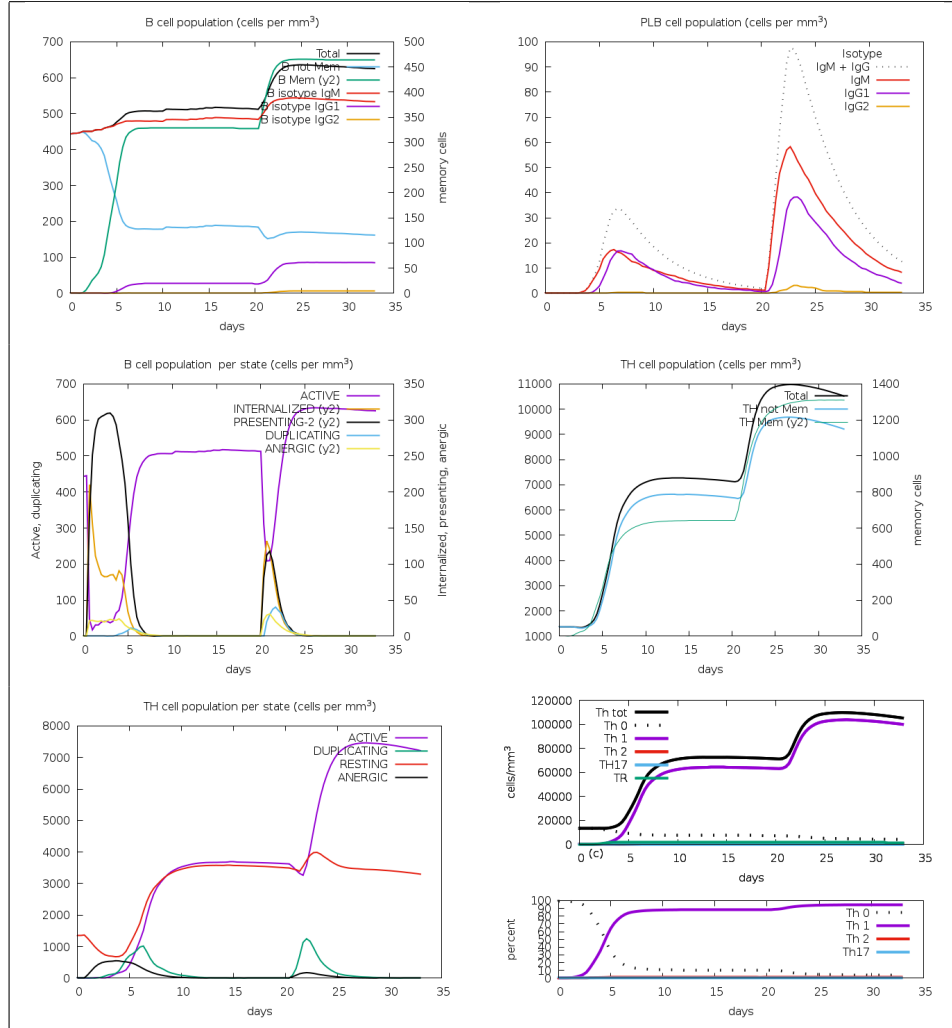


Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

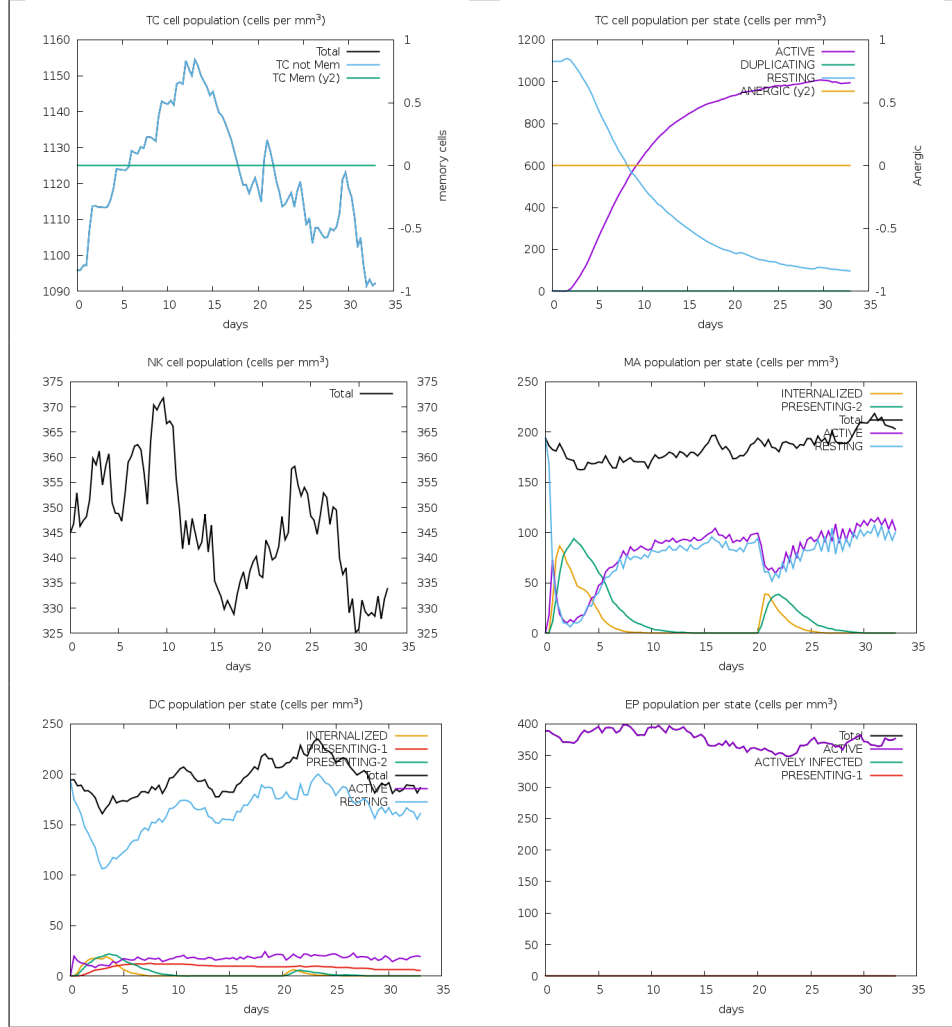


Figure 2: Legend: symbols as figure above.

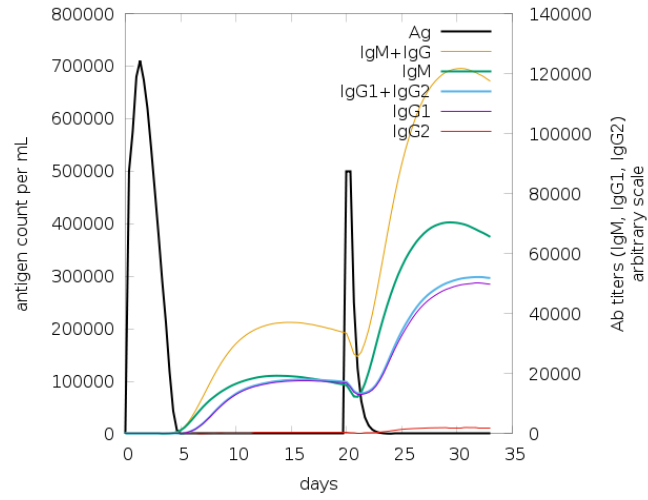


Figure 3: The virus, the immunoglobulins and the immunocomplexes.

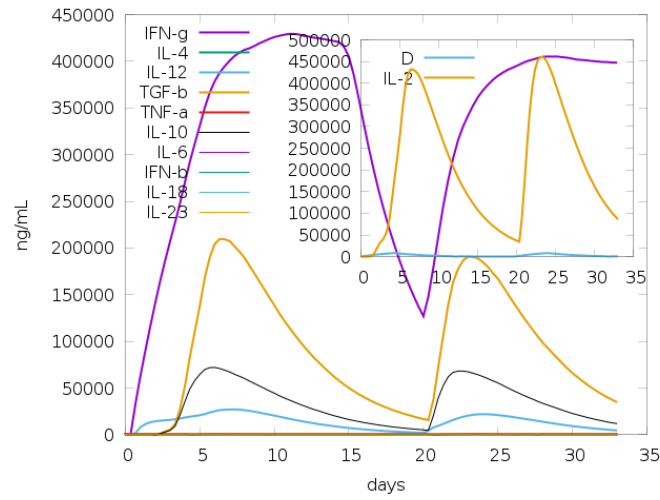


Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

Use Parker's propensity scale, takes an antigen block as input, and creates a list of residues that are possible epitopes.

AKFVAAWTLKAAAGGSANQQQLADIGVPQAPVAAPARRTRKPPQSPESLEECDSELEIKRYKNRVASEAAAKGPGTGP
NGLGEKGDTSPEGSGSGPQRRGGDNHGRGRGRGRGGRRPGAPGSGSGPRHRDGVRRPQKRPSICGCKGTHGGTGA
GAGAGGAGAGGAGAGGAGAGGAGGAGGAGGAGAGGAGAGGAGGAGGAGGAGGAGAGGAGAGGAGAAAKRVLSVVLLGPGPGTM
FEVSVAFGPGPGACMLSAPLEKGPGLPQGGQLTAYGPGPGFLDKGTYTLGPGPGELRLNVTTQGPEAAAKPPMVE

[illegible]

1]	pos=14 len=5	GSANQ
2]	pos=39 len=7	RTRKPQQ
3]	pos=50 len=5	EECDs
4]	pos=73 len=6	GPGTGP
5]	pos=82 len=53	LGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGRRPGAPGGSGSGPRH
6]	pos=136 len=4	DGVR
7]	pos=152 len=66	GTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGGAGAGGGAGAGGGAGAGGGAG
8]	pos=302 len=5	TQGPE

Given the antigen injected creates the list of peptides for all the NumAgProts proteins and for all i.e., 4 MHC I molecules

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

AKFVAAWTLKAAAGGSANQQQLADIGVPQAPVAAPARRTRKPPQPESLEECDSELEIKRYKNRVASEAAAKGPGTGP
NGLGEKGDTSGPEGSGSGPQRRGGDNHGRGRGRGRGGGRPGAPGSGSGPRHRDGVRRPQKRPSICGCKGTHGGTGA
GAGAGGAGAGGAGAGGAGAGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGGAGAGGAGAGGAGAAAKRVLSVVLLGPGPGTM
FEVSVAFGPGPGACMLSAPLEKGPGLPQGQLTAYGPGPGFLDKGTYTLGPGPGELRLNVTTQGPEAAAKPPMVE

```
0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event
```

5

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-020306_5_wd6KTjRQY0VT.FSA_1_001

```
Epitopes of protein 0 -----
0]      pos=  -1 score=1.000000 unnormalised=104.8410000000      non-binding event
```

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-020306_5_wd6KTjRQY0VT.FSA_1_001

```
Epitopes of protein 0 -----
0] pos= 27 score=0.033098 unnormalised=3.8422000000 VPQPAPVAA
1] pos= 29 score=0.031022 unnormalised=3.6012000000 QPAPVAAPA
2] pos= 121 score=0.002595 unnormalised=0.3012000000 RPGAPGGSG
3] pos= 140 score=0.016645 unnormalised=1.9322000000 RPQKRPSCI
4] pos= 264 score=0.013500 unnormalised=1.5672000000 GGPLPQQGL
5] pos= -1 score=0.903140 unnormalised=104.8410000000 non-binding event
```

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-020306_5_wd6KTjRQY0VT.FSA_1_001

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Epitopes of protein 0 -----
0] pos= 27 score=0.033098 unnormalised=3.8422000000 VPQPAPVAA
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```

```
5]      pos=  -1 score=0.903140 unnormalised=104.8410000000      non-binding event
```

DoPeptideList_II:

Given the antigen injected creates the list of peptides for all the NumAgProts proteins and for all i.e., 2 MHCII molecules

Read class II peptide list from file? NO

Allele: DRB1_0101
Pseudo sequence: KAF AHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-020306_5_wd6KTjRQY0VT.FSA_1_001

AKFVAAWTLKAAAGGSANQQQLADIGVPQAPVAAPARRTRKPPQSPESLEECDSELEIKRYKNRVASEAAAKGPGTGP
NGLGEKGDTSGPEGSGGSGPQRGGDNHGRGRGRGRGGGRPGAPGSGSGPRHRDGVRRPQKRPSICGCKGTHGGTGA
GAGAGGAGAGGAGAGGAGAGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAGGAGAGGAGAGGAGAAAKRVLSVVVLLGPGPGTM
FEVSVAFGPGPGACMLSAPLEKGPGLPQGGTLAYGPGPGFLDKGTYTLGPGPGELRLNVTTQGPEAAAKPPMVE

Epitopes of protein 0 -----

0]	pos=	2	score=0.038281	unnormalised=3.3845600000	FVAAWTLKA
1]	pos=	6	score=0.005220	unnormalised=0.4615600000	WTLKAAAGG
2]	pos=	8	score=0.065517	unnormalised=5.7925600000	LKAAAGGSA
3]	pos=	61	score=0.018793	unnormalised=1.6615600000	YKNRVASEA
4]	pos=	65	score=0.009609	unnormalised=0.8495600000	VASEAAAKG
5]	pos=	229	score=0.018929	unnormalised=1.6735600000	VVLLGPGPG
6]	pos=	230	score=0.009632	unnormalised=0.8515600000	VLLGPGPGT
7]	pos=	246	score=0.033135	unnormalised=2.9295600000	FGPGPGACM
8]	pos=	259	score=0.010084	unnormalised=0.8915600000	LEKGPGPGL
9]	pos=	300	score=0.042409	unnormalised=3.7495600000	VTTQGPPEAA
10]	pos=	-1	score=0.748392	unnormalised=66.1680000000	non-binding event

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AKFVAAWTLKAAAGGSANQQQLADIGVPPAPVAAPARRTRKPPQPSLEECDSELEIKRYKNRVASEAAAKGPGTGP
NGLGEKGDTSGPEGSGSGPQRRGGDNHGRGRGRGRGGGRPGAPGSGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGA
GAGAGGAGAGGAGAGGAGAGGGAGGAGGAGAGGGAGAGGGAGGAGGAGGAGGAGGAGGAGGAGAEAAAKRVLSVVVLLGPGPGTM
FEVSVAFGPGPGACMLSAPLEKGPGLPQGGQLTAYGPGPGFLDKGTYTLGPGPGELRLNVTTQGPEAAAKPPMVE

Epitopes of protein 0 -----

0]	pos=	2	score=0.038281	unnormalised=3.3845600000	FVAAWTLKA
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7]	pos= 246	score=0.033135	unnormalised=2.9295600000	FGPGPGACM
8]	pos= 259	score=0.010084	unnormalised=0.8915600000	LEKGP GPGL
9]	pos= 300	score=0.042409	unnormalised=3.7495600000	VTTQGPEAA
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